

Media Access Control Address (MAC)

The MAC address is the physical address of a network interface card (NIC) and is unique.

Consists of six bytes (48 bits).

The first three bytes (24 bits) are assigned by IEEE to the manufacturer, allowing identification of the manufacturer from the MAC address.

Example of a MAC address: 00:21:70:6f:06

or 00-21-70-6F-06-F2

The address FF:FF:FF:FF:FF

is used for broadcasting to all devices.

Internet Protocol (IP)

An IP address is a logical address used to uniquely identify a device on a network.

Two versions exist: IPv4 (Internet Protocol version 4) and IPv6 (Internet Protocol version 6).

IPv4

Representation	1st Octet (8 Bits)	2nd Octet (8 Bits)	3rd Octet (8 Bits)	4th Octet (8 Bits)
Decimal	192.	168.	1.	10
Binary	11000000	10101000	00000001	00001010

Consists of 32 binary bits, divided into four octets.

Each octet is converted to a decimal number and separated by a dot. Example:
192.168.1.10

Each component represents a bit, with each octet consisting of 8 bits.

Each bit is positioned from right to left, increasing in powers of 2.

To convert an octet of 192 to binary, start from the left and add the necessary number of 1s. Each added 1 corresponds to a power of 2. Then add up the values to verify.

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
128	64	32	16	8	4	2	1	
1	1	0	0	0	0	0	0	128+64 = 192

An IP address is divided into two parts:

Network Address

Uniquely identifies each network.

Example: A network created for the accounting department.

Host Address

Uniquely identifies each device within the network.

Example: A specific computer in the accounting department.

IPv4 Address Components

Every device in a network is assigned an IP address, subnet mask, and default gateway:

IP Address: The unique logical address assigned to each device in a network.

Subnet Mask: A value used to determine which subnet an IP address belongs to. It identifies the network portion and the host portion of an IP address. For example, a subnet mask of 255.255.255.0 indicates that the last 8 bits are used for identifying the device while the first 24 bits identify the network.

Default Gateway: The IP address of a router that allows local network devices to communicate with devices on other networks. It handles data transfer between different networks.

In Windows, the `ipconfig` command, and in Linux-based systems, the `ifconfig` command can display the IP address, subnet mask, and default gateway assigned to your computer.

```
[root@hackerbox]~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.20.4.55 netmask 255.255.255.0 broadcast 172.20.4.255
    inet6 fe80::5054:ff:fe0c:f3fd prefixlen 64 scopeid 0x20<link>
    ether 52:54:00:0c:f3:fd txqueuelen 1000 (Ethernet)
    RX packets 29834 bytes 2335990 (2.2 MiB)
    RX errors 3 dropped 0 overruns 0 frame 3
    TX packets 3049 bytes 21113935 (20.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 11 memory 0xfc840000-fc860000

[root@hackerbox]~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.20.5.26 netmask 255.255.255.0 broadcast 172.20.5.255
    inet6 fe80::5054:ff:feab:6324 prefixlen 64 scopeid 0x20<link>
    ether 52:54:00:ab:63:24 txqueuelen 1000 (Ethernet)
    RX packets 22322 bytes 1704341 (1.6 MiB)
    RX errors 2 dropped 0 overruns 0 frame 2
    TX packets 2799 bytes 38108592 (36.3 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 11 memory 0xfc840000-fc860000
```

IPv4 Address Classes

IP addresses are categorized into classes based on network size and address range.

Class	Network Bits	Host Bits	Address Range
A	8	24	1.0.0.0 - 126.255.255.255
B	16	16	128.0.0.0 - 191.255.255.255
C	24	8	192.0.0.0 - 223.255.255.255

Network bits identify the network, while host bits identify devices within that network.

Subnets

Subnets are a network design concept used to divide a large network into smaller and more manageable parts. By logically grouping the devices in the main network, they allow for better management of network traffic and increased network security.

Subnets are typically used to reduce the number of devices on the network, keep network traffic local, and simplify network management. Additionally, subnets can enhance network security by creating firewalls between different departments or functions within the network.

Subnets are defined by a value called a subnet mask. A subnet mask is used to distinguish the network part and the device part of an IP address. For instance, the subnet mask 255.255.255.0 indicates that the last 8 bits identify the device, and the preceding 24 bits identify the network.

In this way, subnets provide network administrators the flexibility to better organize and manage their networks, thereby improving performance and security in the network.

CIDR (Classless Inter-Domain Routing) Notation

CIDR notation specifies the subnet mask more concisely.

192.168.1.0/24 indicates the IP address is 192.168.1.0 with a subnet mask of 255.255.255.0.

/24 = 11111111.11111111.11111111.00000000

/24 = 255.255.255.0

Hosts

To determine the number of hosts in a network, use the formula ($2^{\text{number of host bits}} - 2$).

For an address of 192.168.1.0/24:

Network Bits: 11111111.11111111.11111111.00000000 / 24

Host Bits: 00000000.00000000.00000000.11111111

Calculation: ($2^8 - 2 = 254$) hosts.

Public and Private IP Addresses

Public IP Addresses

Public IP addresses are globally unique addresses assigned by Internet Service Providers (ISPs) and used for internet communication.

Accessible and recognizable by any device on the internet.

Assigned from a global pool managed by international organizations.

Public IP addresses are directly accessible on the internet and are used for public servers, websites, email servers, etc.

Private IP Addresses

Private IP addresses are used within local networks and are not routable on the internet.

Assigned from ranges defined by RFC 1918.

Used in LANs or home networks and require Network Address Translation (NAT) to communicate with the internet.

Type	Start IP Address	End IP Address	Subnet Mask
Public	0.0.0.0	126.255.255.255	0.0.0.0/1
	128.0.0.0	191.255.255.255	128.0.0.0/2
	192.0.0.0	223.255.255.255	192.0.0.0/3
	224.0.0.0	239.255.255.255	224.0.0.0/4
	240.0.0.0	255.255.255.254	240.0.0.0/4
Private	10.0.0.0	10.255.255.255	10.0.0.0/8
	172.16.0.0	172.31.255.255	172.16.0.0/12
	192.168.0.0	192.168.255.255	192.168.0.0/16
	169.254.0.0	169.254.255.255	169.254.0.0/16

The Loopback Address

The loopback address (generally 127.0.0.1), used in computer networks, is a special IP address that routes the data back to the same device. The loopback address is used for a computer to communicate with itself and test network applications.

For example, if a web developer runs a local web server and wants to access it via a browser, they can use <http://127.0.0.1> or <http://localhost>. This allows the developer to test the web application on their local machine.